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Fundamentals of Energy Analysis for Crop Production Agriculture

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Energy is embodied in all of the equipment, inputs, and products of agriculture. Agriculture both uses and supplies energy in the form of bioenergy and food. The amount of energy used in agriculture has grown substantially, and currently, the agri-food chain accounts for 30 percent of the total energy used around the world [1]. Sustainable agricultural production requires the optimization of land use, energy efficiency, end of the use of fossil energy sources, and minimization of environmental impacts. Current agricultural systems are heavily dependent on fossil energy resources. Energy analysis allows the quantification of the amounts of energy used for agricultural production and can be used to optimize energy consumption and increase energy efficiency to move agriculture closer towards sustainability.

Energy Analysis Methodology in Production Agriculture

STEP 1: DEFINING SCOPE AND FUNCTIONAL UNIT

The scope of energy analysis depends on the objective. Usually, the scope of energy analysis in production agriculture is confined to activities required for agricultural production, from the manufacture of inputs, energy used for production, and energy for storage of the products before they are sold (Figure 1). Energy consumption for different activities within an agricultural production system are usually defined in relation to the functional unit. Thus, the functional unit is usually expressed in terms of mass (e.g., 1 ton), area (e.g., 1 acre), or economic value (e.g., $1000 revenue). For estimating the

overall energy requirements for different activities related to an agricultural production system, the direct inputs to all the activities (Figure 1) are identified. However, agricultural operations vary in their energy use based on the details of the production system. For example, a zero-till rain-fed corn production system may have much different energy use balance than an irrigated tillage system.

STEP 2: QUANTIFYING INPUTS TO DIFFERENT ACTIVITIES OF THE AGRICULTURAL PRODUCTION SYSTEM

In energy analysis, energy and material requirements are estimated for the manufacture and transportation of inputs used for the different agricultural activities considered in the analysis. This allows the quantification of the amount of energy used for production, harvest, and post-harvest logistics operations in terms of the same functional unit. Figure 1 shows the energy using activities that must be accounted for in energy analysis for crop production agriculture. Inputs for crop production, harvest, and post-harvest logistics operations may include seeds, chemical fertilizers (nitrogen, phosphate, potassium, and sulphur), pesticides (herbicides, fungicides, and insecticides), diesel fuel, lubricants, electricity, manure, irrigation water, labor, and machinery. Some operations require special machines and equipment, such as tractors, cultivators, spreaders, sprayers, harvesters, irrigation systems, transportation equipment, and dryers, as well as facilities, such as transportation terminals and storage facilities for agricultural products and by-products.

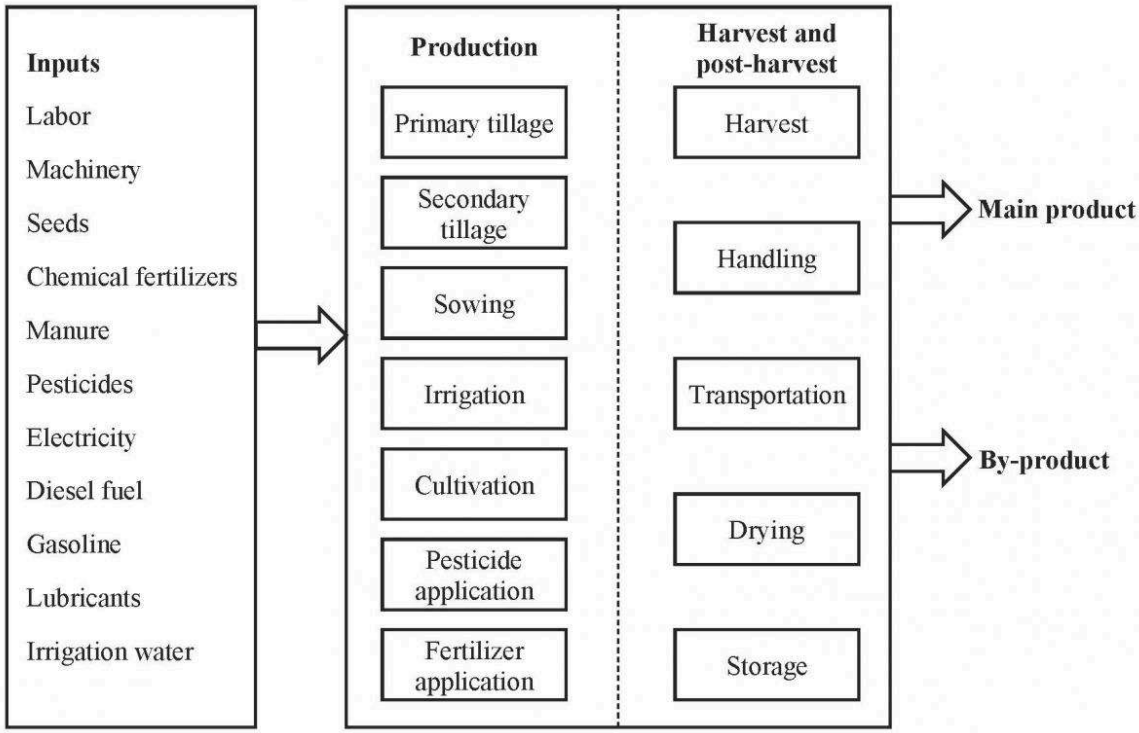

Figure 1. Farm practices and inputs for crop production agriculture

STEP 3: CONVERTING THE PHYSICAL INPUTS AND OUTPUTS TO ENERGY

The physical inputs and outputs of different agricultural operations need to be converted to common energy units using energy equivalent coefficients (Table 1). The energy equivalent coefficient of an input is defined as the sum of the energy consumed during the production of the input and the energy used for transportation of the input to the end user or local market. The energy used for production and transportation logistics varies based on differences in technology in different regions and countries, energy consumption for the specific production process used, and transportation distance and technology. So, the energy equivalent coefficient of an input varies substantially as a function of location. Energy equivalent coefficients for agricultural inputs are presented in Table 1.

Table 1: Energy equivalent coefficients for production agriculture inputs.

Input	Unit	Energy equivalent coefficients ($\times 10^3$ BTU Unit^{-1}) [Reference]

1. Labor	h	1.50 [2], 1.86 [3]
2. Machinery	lb	
a. Tractor		40.25 [4]
b. Self-propelled combine		37.67 [4]
c. Agricultural equipment		27.0 [4]
3. Diesel fuel	gal	171.50 [5], 202.03 [6]
4. Chemicals	lb	
a. Herbicides		101.89 [2], 102.32 [3], 126.83 [7]
b. Insecticides		24.94 [7], 43.51 [3]
c. Fungicides		49.44 [7], 84.26 [2], 92.86 [3]
5. Fertilizer	lb	
a. Nitrogen		21.50 [2], 28.44 [8], 33.58 [5]
b. Phosphate (P_2O_5)		5.09 [2], 5.35 [8], 7.48 [5]
c. Potassium (K_2O)		1.81 [2], 4.79 [8], 5.89 [5]
d. Sulfur (S)		0.48 [8]
e. Manure		0.13 [8]
6. Irrigation water	ft^3	0.03 [8]
7. Electricity	kWh	3.41 [9], 11.31 [8], 11.85 [2]

8. Seeds	lb	10.75 for cereals and pulses [7], 43.0 [10] and 44.5 [11] for corn grain, 1.55 for oilseeds [7]

The energy embodied in the production of farm machinery and tractors is assumed to be depreciated over the economic lifetime of the equipment [12]. The depreciated energy expressed as the fractional weight of the equipment for a working unit of a specific process is estimated using expression (1), as suggested by reference [13]. The energy associated with the machinery is then calculated by multiplying the energy equivalent coefficient of the machinery (in BTU per pound) by the depreciated weight (in pounds).

$$\text{Depreciated weight [lb]} = \text{Machine weight [lb]} \times \frac{\text{Operation time [h]}}{\text{Economic lifetime [h]}} \quad (1)$$

For example, the depreciated weight for 10 hours of field work by an 8000-pound tractor with an economic lifetime of 12,000 hours is calculated as:

$$8000\ [\text{lb}] \times \frac{10\ [\text{h}]}{12{,}000\ [\text{h}]} = 6.67\ [\text{lb}] \quad (2)$$

Irrigation is a major energy using process in arid regions of the world. Energy for water pumping alone may be several times greater than that for all the other agricultural field operations combined [14]. Energy used for irrigation is both direct and indirect. Direct energy for irrigation is mostly consumed as electricity, diesel fuel, and labor. Indirect energy for irrigation consists of the energy consumed for manufacturing the materials for dams, canals, pipes, pumps, and equipment, as well as for construction of the works and buildings used for the on-farm irrigation system [14]. Sunlight energy used for crop photosynthesis is not quantified in energy analysis since there is no cost for this type of energy [14].

Energy consumption for each agricultural operation is estimated by summing the energy consumed for each of the different inputs ed for that operation. Energy consumption for different operations contributes towards the “total energy input” for the agricultural production system. The magnitude of energy consumption from different operations, or production systems, can then be compared based on the pre-defined common functional unit.

STEP 4: INTERPRETING THE RESULTS OF ENERGY ANALYSIS

As a final step, the energy ratio (Eq. 3) is determined in which “energy output” (i.e., stored chemical energy of the product) is assessed relative to “total energy input” [15]. The other indicator is energy productivity, which quantifies the production rate per unit energy use (Eq. 4). The net energy return is the absolute difference between “energy output” and “total energy input” (Eq. 5).

$$\text{Energy ratio} = \frac{\text{Energy output [BTU]}}{\text{Total energy input [BTU]}} \quad (3)$$

$$\text{Energy productivity [lb/BTU]} = \frac{\text{Production mass [lb]}}{\text{Total energy input [BTU]}} \quad (4)$$

$$\text{Net energy [BTU]} = \text{Energy output [BTU]} - \text{Total energy input [BTU]} \quad (5)$$

Some other energy indicators focus on the classification of energy consumption based on energy sources and types. The energy requirement for agricultural production is classified as direct and indirect, as well as renewable and non-renewable. Direct energy inputs include those that are directly consumed during crop production, harvest, and post-harvest operations. These include actual energy contained in diesel fuel, electricity, and labor. Indirect energy includes sequestered energy in seeds, farmyard manure, chemical fertilizers, pesticides, irrigation equipment, and all types of machinery. Non-renewable energy sources include diesel fuel, pesticides, chemical fertilizers, electricity made from coal or natural gas, and machinery, while renewable energy sources, in general terms, consist of labor, seeds, and farmyard manure, and any type of renewable energy that is used during the operations [16].

Conclusions

Energy use analysis in production agriculture is essential for development of more efficient production systems. Energy is either directly or indirectly consumed in different operations of agricultural production, including cultivation, harvest and post-harvest logistics. Energy analysis of agricultural production systems is conducted by defining the goal and scope, assessing input and output parameters, assigning their energy equivalents, and quantifying energy use indicators. Interpretation of the results can help policymakers, farmers, and manufacturers improve energy efficiency and the sustainability of production agriculture.

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